



# MEHADI HASAN

AI / LLM SOLUTIONS ARCHITECT

Senior Software Engineer · Tech Lead

## CONTACT

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## AI / LLM STACK

LLM & Agentic AI Systems

RAG / Vector Search

LangChain / LangGraph

AWS Bedrock / Azure OpenAI

n8n / Workflow Automation

Prompt Engineering / Evals

## TECH SKILLS

HTML / CSS / JavaScript

TypeScript / React / Vue

Python / C / C++

Node.js / NestJS

Java / Kotlin / Go

Flutter / Dart

ASP.NET / C# / WPF

PHP / Laravel

REST / gRPC / GraphQL

Kafka / RabbitMQ / MQTT

## SOFT SKILLS

Leadership

Team Management

Problem Solving

## LANGUAGES

- English
- Bangla
- Hindi
- Malay (Basic)

## 14+ Years · Government · Fintech · Healthcare · Enterprise

Production GenAI & Agentic AI · Mission-Critical Systems · Multi-Team Leadership

## PROFESSIONAL SUMMARY

Results-driven technologist with **14+ years** delivering mission-critical systems across **government, fintech, healthcare** and **enterprise** domains. Architect of production **Generative AI & Agentic AI** platforms – **LLM integration** (OpenAI, Anthropic Claude, Google Gemini, Ollama), **RAG pipelines**, **LangChain / LangGraph** orchestration, and **Vector Databases** (Pinecone, Weaviate, Qdrant, Chroma, FAISS, Milvus, pgvector) on **AWS Bedrock** and **Azure OpenAI**. Power user of **AI pair programming** (Claude Code, GitHub Copilot, Gemini Code Assist, OpenAI Codex). Deep expertise in **microservices**, **system design (HLD/LLD)**, **event-driven architecture** (Kafka, RabbitMQ), **n8n automation**, and **MLOps / LLMOps**. Currently **Senior Engineer & Lead** at GIS, owning architecture for 3 Malaysia Government transport & infrastructure programmes. Pursuing **PhD in Computer Science**.

## CORE COMPETENCIES

LLM Integration Agentic AI RAG Vector Search LangChain LangGraph LlamaIndex AWS Bedrock Azure OpenAI Vertex AI MCP Servers Prompt Engineering LLM Evals MLOps / LLMOps n8n HLD/LLD Microservices Event-Driven Kafka RabbitMQ Real-time Cloud Native CI/CD OWASP Top AI Security Open Source

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer & Lead

Sep 2024 – Present

GIS – Government Projects: LRT3, RTS, Smart Car Parking · Malaysia

- Lead engineer across **3 simultaneous Malaysia Government programmes**; own end-to-end HLD/LLD for **4-team, 18-engineer** organisation
- **LRT3**: C# .NET + PostgreSQL signalling backend processing **>2M events/day**; cut alarm noise **~60%** via custom IP Scanner with multicast tracing
- **Smart Car Parking (TNG/Genting)**: Flutter/Dart + NestJS + AI OCR; reduced gate latency from **4s → 800ms** on integrated TNG/Mifare payment flow
- **MyDesk**: Custom AnyDesk-alternative remote desktop unifying control of all 3 programmes; eliminated 3 commercial-licence costs (**~RM 60k/yr saved**)
- Built **Kafka TUI**, **RabbitMQ TUI** now standard diagnostic for the LRT3 ops team
- Introduced **LLM-assisted log triage** (Claude/OpenAI) + **n8n** ops pipelines; cut mean-time-to-diagnose by **~40%**
- SCADA via JMS; CISCO VMware ESXi; **Grafana** monitoring across 3 programmes

### Engineering Lead (Contract, Remote)

May – Oct 2025

Slotgator iGaming Project · Romania

- Delivered iGaming platform on **AWS + Kubernetes** sustaining **~5k concurrent sessions**; zero P0 incidents over 6-month engagement
- Drove **Kafka** streaming + microservices, AES-256 payment hardening; led 7-engineer distributed team across 4 time zones

### Full Stack Developer & DevOps Engineer

Dec 2023 – Sep 2024

Gaming Industry · Malaysia & USA

- Shipped 4 hybrid iOS/Android apps reaching **>100k installs**; managed **AWS EC2**, Tencent & Alibaba Cloud infra
- Integrated 3 payment rails (Decoin, Alipay, UnionPay) with AES-256 + real-time Web-Socket; CI/CD cut release cycle from 2w → 2 days

### Project Manager / Team Lead

Dec 2022 – Oct 2023

Creative IT Institute · Chittagong

- Led **12-engineer** cross-functional team across 5 client products using Agile/Scrum; delivered **100%** on-schedule release rate
- Designed PostgreSQL/MySQL schemas powering **~30k DAU**; standardised Git, VPS & Nginx operating procedures

### Team Development Lead & Sr. Full Stack

Mar – Oct 2022

Datahostbd · Dhaka

- Mentored **8 junior engineers**; authored microservices training manual still in use today
- Architected microservices with API gateway + service discovery (SOLID); reduced cross-service latency **~35%** vs. monolith baseline

KEY INTERESTS

- ◊ Large Language Models
- ◊ Agentic AI & Autonomous Agents
- ◊ Generative AI Applications
- ◊ RAG & Vector Search
- ◊ LangChain / LangGraph
- ◊ AWS Bedrock / Azure OpenAI
- ◊ Model Context Protocol (MCP)
- ◊ n8n Workflow Automation
- ◊ AI in Healthcare & EdTech
- ◊ MLOps & LLMops
- ◊ Prompt Engineering & Evals
- ◊ Microservices Architecture
- ◊ Real-time Systems
- ◊ Cloud Native Development
- ◊ Cyber Security & R&D
- ◊ IoT & Embedded Systems

EXPERTISE

- ✓ AI / LLM System Design
- ✓ Agentic AI Workflows
- ✓ RAG Pipeline Engineering
- ✓ Full Stack Architecture
- ✓ System Design (HLD/LLD)
- ✓ Microservices
- ✓ Event-Driven Architecture
- ✓ API Development
- ✓ Cloud Infrastructure
- ✓ Real-time Systems
- ✓ CI/CD Pipelines
- ✓ Agile / Scrum Leadership
- ✓ Technical Consulting
- ✓ Team Mentorship
- ✓ Performance Optimization

COMPETENCIES

- LLM Integration
- Multi-Agent Orchestration
- Retrieval-Augmented Generation
- AI Pair Programming
- Machine Learning
- Distributed Systems
- Neural Networks

PLATFORMS

Unix, Linux, Windows, Android, iOS, macOS, AWS (Bedrock, EC2, S3, Lambda), Azure (OpenAI), GCP (Vertex AI), Docker, Kubernetes, VMware ESXi

PROFESSIONAL EXPERIENCE (CONT.)

Senior Consultant & Full Stack Developer

Mar 2020 – Jan 2022

Creative IT Institute · Chittagong

- Hardened 6 production codebases through code review + unit-test rollout; reduced post-release defects ~**50% YoY**
- Coordinated large-scale deployments serving ~**8k students**; co-designed REST APIs with backend engineers

Chief Consultant & Engineer

Jan 2017 – Aug 2019

TRZ · Gazipur

- Built end-to-end ERP for garments industry from scratch; RFID-integrated tracking across **4 factory floors, 600+ daily operators**
- Automated scheduled power management saving ~**18%** electricity cost; BI model in Matplotlib/Pandas/NumPy adopted by exec leadership

CEO & Chief Consultant

Jul 2011 – Nov 2016

Orimpo Inc. Ltd · Dhaka

- Founded & ran consultancy for 5+ years; delivered **40+ enterprise projects** on J2EE/JSP/To across govt, fintech, retail
- Led **bKash mobile payment prototyping** with Robi Axiata (Java) – foundational work for Bangladesh’s largest mobile-money platform
- Conducted security assessments & network penetration testing for govt & enterprise clients

EDUCATION

Doctor of Philosophy (PhD) – Computer Science

Jul 2025 – Ongoing

Lincoln University College · Part-time

Master of Computer Science

Sep 2023 – May 2025

Lincoln University College

Thesis: AI-Driven Medical Assistant

BSc in Computer Science & Engineering

2009 – 2017

United International University

Project: An Intelligent Building Power Management System (IoT)

Coursework: Java, PHP, Python, .Net, J2EE, ML, Neural Networks, Data Mining. Awards in Programming Contest & Robotics.

ADDITIONAL CONSULTANCY

- ▶ **Alamin Sweaters** (Feb – May 2013) – Technical consulting for garments manufacturing, production workflow optimization, barcode-based material tracking
- ▶ **Friends Appeals** (Mar – Aug 2018) – Custom business management software, database architecture, reporting systems
- ▶ **Peregrine Air Line (UK)** (Jan – Jun 2018) – E-commerce platform with stock management, shipment tracking, barcodes, payment gateway

SPEAKING, MENTORSHIP & COMMUNITY

- Author of open-source **Redmine plugin suite** (6 plugins) used by self-hosted teams worldwide for project management UX improvements
- Mentored junior developers across Bangladesh & Malaysia teams; produced internal training manuals on microservices, SOLID principles & secure coding
- Active contributor to **Generative AI / Agentic AI** community via public repositories (SuperCharge Build Challenge, Free LLM API Keys aggregator, KafkaTUI, RabbitMQ-TUI)
- Internal knowledge-sharing sessions on **AI-assisted software engineering** & mission-critical signalling architecture for engineering teams across Malaysia Government transport programmes

METHODOLOGY

- Scrum / Agile / Waterfall
- SOLID Principles
- Domain-Driven Design
- Redmine / Jira
- Git / Bitbucket
- Jenkins CI/CD
- Test-Driven Development
- LLM Evals & Red-Teaming

DATABASES

**Relational:** Oracle, MSSQL, MySQL, PostgreSQL, SQLite  
**NoSQL:** MongoDB, Redis, Firebase  
**Vector:** Pinecone, Weaviate, Qdrant, Chroma, FAISS, Milvus, pgvector

AI / GENAI STACK

**LLM APIs:** OpenAI, Claude, Gemini, Grok, Mistral  
**Self-Hosted:** Ollama, vLLM, llama.cpp  
**Orchestration:** LangChain, LangGraph, LlamaIndex  
**Agents:** MCP, Function Calling, ReAct, AutoGen  
**Cloud LLM:** AWS Bedrock, Azure OpenAI, Vertex AI  
**Eval:** LangSmith, Langfuse, Promptfoo, RAGAS

DEV TOOLS

VS Code, Neovim, JetBrains, OWASP ZAP, n8n, Grafana, RabbitMQ, Ollama, Claude Code, GitHub Copilot, Gemini Code Assist, OpenAI Codex, LaTeX, Overleaf  
**Custom Tools:** MetaSig, KafkaTUI, RabbitMQTUI, IPScanner (custom network mapping & visualization), custom ticket monitoring  
**Networking:** Tailscale VPN, OpenVPN, WireGuard, multicast tracing

PORTFOLIO VIDEOS

- ▷ LRT / MRT / RTS Signaling
- ▷ Project Management
- ▷ Inventory System
- ▷ UK Export / Import
- ▷ Smart Building IoT
- ▷ Gaming Project
- ▷ iGaming Project

SELECTED PROJECTS

- **MetaSig – Cross-Platform Remote Device Management Platform** (2025 – Present) – Production fleet-management TUI for Android/iOS/macOS/Win/Linux: WebSocket signalling, remote shell, file browser, live screen preview, APK packaging & signing pipeline. Manages >200 endpoints across 3 Malaysia Gov programmes from a single terminal. *Stack:* Node.js, neo-blessed TUI, WebSocket, Android agent
- **GPT-5.3 Agent Bridge + Pentest-AI Red-Team Deployment** (2025 – Present) – Built OpenAI-compatible HTTP bridge wrapping ChatGPT GPT-5.3 via Playwright; integrated as multi-model agent inside **opencode** TUI for agentic dev workflows. *Separately:* authorised internal red-team deployment of upstream open-source Pentest-AI (**ptai**, original by 0xSteph) for OWASP Top 10 exercises via MCP – *integration role, not authorship*
- **Smart Car Parking System (TNG / Genting)** (Sep 2024 – Present) – AI OCR plate-detection + C/C++ hardware gate control + TNG/Mifare payment integration; cut gate-pass latency from 4s → 800ms. *Stack:* Flutter/Dart, NestJS, C# WinForms (CPMS SAS Scheduler), Python OCR, PostgreSQL
- **LRT3 / RTS Signalling Systems + KafkaTUI / RabbitMQ-TUI** (Sep 2024 – Present) – Mission-critical Malaysia Gov. transport signalling processing >2M events/day (C# .NET + PostgreSQL + Java/PHP). Open-source TUI diagnostics now standard for LRT3 ops team; cut alarm noise ~60%
- **Medical AI Assistant (MSc Thesis) + AI Doctor Triage Platform** (2024 – 2025) – HIPAA-aware healthcare RAG over clinical knowledge base (Claude / OpenAI / Ollama, LangChain, LangGraph, Pinecone/Qdrant/pgvector, FastAPI). AI Doctor: Gemini-powered triage with adaptive symptom intake; React Native + Node.js + Prisma; Dockerised
- **Open-Source Suite – 16+ public repos** (2023 – 2026) – **mydesk** (AnyDesk-alternative), **macscp** (Swift SFTP), **snmp-alarm** (NOC alarming), **SignalR→WS bridge**, **nginx-RTMP for macOS**, **SuperCharge SG** (Telegram RAG bot), **Free LLM API Keys**, 6 Redmine plugins (**forum-pdf**, **doc-pdf**, **turbo-nav**, **issue-emailer**, **dark-mode**, **custom-mail**)
- **Earlier flagship work** (2015 – 2024) – P2P Remote Desktop (C++/Qt + WebRTC, H.264/VP8); Custom Dev TUI Suite (multi-framework scaffolder, audio editor, DB exporter, .deb packager); Job-Search AI (9-source aggregator, AI cover letters, SMTP apply); n8n / Ollama infra; Telegram Dating Bot (Stripe + Drizzle); Hathe Khori (TF/Flutter); DMP crime clustering; RFID ERP + encrypted IRC for Bangladesh Army

CERTIFICATIONS & ACHIEVEMENTS

|                                                                                     |                                                                                      |                                                                                       |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |
| Master of Computer Science<br>Lincoln Univ. – 2025                                  | Microsoft Imagine Cup<br>2012                                                        | Programming Contest<br>Qazi Azher Ali – 2011                                          |
|  |  |  |
| UIU Robotics Workshop<br>2011                                                       | NYDSA Certificate<br>National Youth Dev.                                             | Code Warriors<br>Programming Comp.                                                    |

REDMINE PLUGIN SUITE (OPEN SOURCE)

Six published Redmine plugins on [github.com/eshad](https://github.com/eshad): **forum-pdf-email**, **document-pdf-email**, **turbo-nav** (AJAX navigation), **issue-emailer**, **dark-mode**, **custom-mail**.

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